

Every Asset Its Own Benchmark: Market-Neutral Alpha in Perpetual Futures

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ABSTRACT

Cross-sectional factor investing assumes assets can meaningfully be ranked against one another. Equities supply the anchor that makes this reasonable: shared accounting, industry structure, cash flow. Cryptocurrency perpetual futures supply none of it. We ask what happens to factor construction without it. Transporting ten factors from the published equity literature to 112 perpetual contracts over a 363-week panel, we find that ranking each contract against its own trailing history dominates ranking it against its peers in nine of eleven cases, raising mean annualised Sharpe from 1.01 to 1.45; the effect persists with the carry component removed entirely. Standardising each asset against its own history before ranking cross-sectionally recovers 41% of the gap, identifying scale heterogeneity as one channel. We further show that preserving factor independence through construction, by forming one book per factor and combining returns rather than scores, adds 0.665 Sharpe ($t = 5.82$) when both constructions are tuned over an identical grid. Regressed on market, size and momentum factors the strategy retains an annualised alpha of 27%, with a market loading indistinguishable from zero. Through the 2022 decline, in which the market fell 82%, it returned +9% with a maximum drawdown of 8%. We introduce a factor exploiting aggressor-side information that centralised venues publish exactly and equity research must infer. Results survive walk-forward selection across seven rebalance anchors, a held-out universe test, costs at four times the modelled level, and a deflated Sharpe correction whose trial count is read from an executed log.

Keywords: factor investing, cross-sectional ranking, cryptocurrency, perpetual futures, portfolio construction, backtest overfitting

JEL: G11, G12, G14, C58

Disclosure. AI tools were used in this research and for re-verification of the analysis code and tests. All results were independently verified by re-execution, and the author is responsible for all claims and interpretations.

1 Introduction

Every cross-sectional factor model begins with a comparison. We say an asset is cheap, or small, or heavily traded, and each term carries an implicit completion: *relative to its peers*. The comparison is defensible in equities because the market supplies scaffolding that makes it meaningful. Firms report under common accounting standards, occupy classifiable industries, and are valued against cash flows that exist. Ranking them compares like with like.

Cryptocurrency perpetual futures dissolve that scaffolding. There are no earnings, no industry classification, and no accounting identity tying price to anything. Whether one contract’s turnover is high *relative to another’s* is a question the data cannot really answer, because the quantities were never commensurate. Yet the factor literature, in turning to this market, has largely imported the cross-sectional apparatus unchanged.

This paper asks what that assumption costs, and what replaces it.

Our design is deliberately conservative. Rather than proposing new signals, we transport ten factors established in the published equity literature to 112 perpetual contracts across a 363-week panel, retaining their original definitions and sign conventions. Nothing is estimated. The only quantity we vary is the frame of comparison: in one specification each contract is ranked against its peers within the week, as the literature does; in the other, against its own trailing history, asking not whether an asset is unusual relative to others but whether it is unusual *for itself*.

The eleven factors are the *sample*, not the contribution. A claim about construction demonstrated on a single factor would be a curiosity, indistinguishable from a fortunate choice of signal; demonstrated across eleven factors drawn from the published literature and spanning price, flow, positioning and trade-size distribution, it is evidence about the method rather than about any particular signal. We accordingly report per-factor results throughout and never a single headline attributable to one of them.

First, the comparison frame matters. Ranking each contract against its own history outperforms ranking it against peers for nine of the eleven factors we examine, improving mean risk-adjusted performance by roughly forty per cent. The advantage is not an artefact of the funding component, which we find to be a first-order signal in its own right: repeating the comparison with funding removed entirely leaves the ordering intact.

Second, we identify a mechanism rather than only an effect. An intermediate specification, standardising each asset against its own history before ranking cross-sectionally, recovers roughly two-fifths of the advantage, identifying scale heterogeneity as one channel. The remainder survives standardisation, and we attribute it to distributional robustness: a rank transform is invariant to the shape of the distribution it is applied to, whereas standardisation normalises only the first two moments and implicitly assumes approximate normality. Crypto factor distributions violate that assumption severely.

Third, construction dominates selection. Standard practice blends factor scores into a composite and forms one portfolio from the induced ordering. This compresses several near-independent signals into a single ranking and discards their independence before any position is taken. Forming one book per factor and combining the resulting *returns* instead improves performance substantially and reliably, favourable in fifteen of seventeen matched comparisons. Both constructions were tuned over an identical grid; comparing a tuned method against an untuned baseline is the most common way this literature manufactures a result, and we take care not to.

Fourth, we exhibit a factor whose construction depends on information this market publishes and equity research must infer. Work on Chinese A-shares established that predictive content resides in the *shape* of the per-bar trade-size distribution rather than its mean. That measure is necessarily direction-blind, since A-share tick data does not reliably identify the initiating side. Perpetual futures publish taker-side volume on every bar, permitting the distribution to be partitioned by initiating side and its asymmetry measured. The economic motivation is Kyle (1985): an informed trader must transact in size to profit, and trades off concealment against execution speed, leaving a detectable footprint rather than none. We do not overstate the novelty: US equity researchers can infer direction via the tick rule (Lee and Ready 1991), with known error rates. The distinction is measured versus estimated. The resulting factor ranks third of the eleven by its marginal contribution to the portfolio.

Throughout we hold ourselves to the significance standard of Harvey et al. (2016), who argue that the proliferation of published factors implies a hurdle of $t > 3.0$ rather than the conventional $t > 2.0$. We report where we clear it and where we do not. Appendix A summarises every statistical test applied in this paper and its outcome; the tests themselves are developed in Sections 5–7.

Finally, this paper reports an audit of its own claims. During preparation we searched the research codebase for the executable artifact behind every headline figure; four had none, including a pair of statistics that proved to be tail-index values transcribed from a document about an unrelated strategy. Section 8 reports this alongside the infrastructure now preventing recurrence. We regard the reliability of a reported backtest as an empirical property of the pipeline producing it, and therefore as something to be tested.

Three elements of this study appear to be without precedent in the published literature, and we state them as such while noting that we have not conducted an exhaustive search.

First, we are not aware of prior work that uses the *skewness of the trade-size distribution, measured separately by initiating side and differenced over time*, as a cross-sectional return predictor. Two adjacent literatures come close without meeting. Zhou (2008) partitions the size distribution by side in an order-driven equity market, but measures *tail exponents* and finds them statistically indistinguishable across sides ($\alpha = 2.30$ versus 2.36). Separately, a large literature prices the skewness of *returns*. Neither measures the skewness of *trade sizes*, and neither examines how its between-side asymmetry evolves.

The distinction matters for interpretation as well as for priority. Zhou’s null result says the two sides’ size distributions are equally fat-tailed; our result says their *shapes shift at different times*, and that those shifts predict. The two are consistent: a distribution can have a stable tail exponent while its skewness moves.

Second, the requirement that the deflation inputs be measured rather than assumed is understood in principle, the dispersion term cannot be formed without the full set of trial Sharpes, but it is rarely satisfied in practice, because the trial set is not usually retained. We report a study in which both inputs are read from an append-only log written by the code performing each evaluation, and we report the resulting value under two defensible definitions of the trial set, including one that does not clear the threshold.

Third, we are not aware of a published strategy paper that verifies the deployed system computes the same positions as the research code. We report an automated equivalence gate that does so, and we report that it previously *failed*; research and production agreed perfectly while both called an incorrect function. The claim that a live system implements a published backtest is, on our evidence, an empirical proposition rather than a safe assumption.

We also combine two robustness procedures that to our knowledge have not been applied together: walk-forward configuration selection and rebalance-timing-luck variation. Applied separately, each leaves a channel open, a walk-forward result still inherits the arbitrary choice of rebalance weekday, and a timing-luck study still inherits a fixed configuration. Section 6.1 closes both simultaneously.

The rest of the paper is organised as follows. Section 2 places the study in the literature. Section 3 describes the data and the two properties of the venue that make the exercise possible. Section 4 sets out the ranking primitives, portfolio formation and cost model. Section 5 reports the main results. Section 6 subjects them to five independent tests. Section 7 estimates capacity. Section 8 reports the claims that did not survive our own audit, and Section 9 concludes.

2 Related literature

Cross-sectional factor construction. The characteristic-sorted portfolio is the workhorse of empirical asset pricing (Fama and French 1993), and the debate over whether characteristics or covariances drive returns (Daniel and Titman 1997) presumes throughout that assets are commensurate enough to sort. We take that presumption as the object of study rather than as background.

Factor transport across asset classes. Asness et al. (2013) show that value and momentum earn premia consistently across eight markets and asset classes, establishing that factor logic travels beyond the equities in which it was discovered. We transport factors in the same spirit, but ask a question their design does not: whether the cross-sectional *frame* travels with them.

Why we rank rather than forecast. Gu et al. (2020) document that single-asset return prediction is a low signal-to-noise problem even with flexible models. Our book never forecasts a return; it ranks contracts and holds a dollar-neutral spread, so only the ordering must be correct.

Time-series versus cross-sectional signals. Moskowitz et al. (2012) established that an asset’s own past return predicts its future return independently of any cross-sectional ordering. Our ranking primitive is the same statistical object; the contribution is not the primitive but its systematic application across eleven factors spanning price, flow, positioning and trade-size distribution, with a decomposition of why it works.

Cryptocurrency asset pricing. Cong et al. (2026) construct a four-factor cross-sectional model for digital assets, introducing a value factor built from the ratio of active addresses to market capitalisation alongside market, size and momentum factors, and document pronounced segmentation across token categories. Their design is instructive for our purposes in two ways. First, it is *cross-sectional*, which makes Section 5.1 a direct empirical response rather than a parallel exercise: we find that the same ranking frame they employ is materially weaker in this market than ranking each contract against its own history. Second, and more interestingly, their value factor is an attempt to *manufacture* the comparability anchor that equities possess natively, active addresses stand in for the fundamental quantity crypto does not report. That such a construction is necessary is itself evidence for the problem we identify. Our response differs in kind: rather than synthesising an anchor, we abandon cross-asset comparison where it is not warranted. The two approaches are complementary, and we regard testing their factors under a self-referential frame as the natural next step.

Backtest overfitting. Bailey and López de Prado (2014) formalise deflation of an observed Sharpe by the expected maximum across trials; Harvey et al. (2016) raise the significance hurdle; Hoffstein et al. (2019) document that rebalance-schedule choice alone generates dispersion often exceeding the effect under study. We apply all three, and in the first case address the standard weakness that the trial count is supplied by the researcher.

3 Data

Universe. 112 USDT-margined perpetual futures contracts, screened for continuous trading and finite positive prices, spanning 363 weeks. After factor warm-up the strategy is evaluable on 308 weekly rebalances and active on 282 (Table 1).

Sources. Price and volume derive from hourly bars in a public bulk archive, resampled weekly. Intraday inputs are reduced from 1-hour and 5-minute bars to one record per UTC day. Positioning data, open interest, top-trader and all-account long/short ratios, comes from a keyless exchange metrics endpoint at daily frequency. No proprietary or paid data is used.

Two venue properties are load-bearing. First, aggressor side is published: every bar carries taker-buy volume, where equity microstructure research must infer direction (Lee and Ready 1991).

Table 1: Sample and universe.

Universe	112 USDT-margined perpetual futures
Panel	363 weeks
Rebalances evaluable / active	308 / 282
Rebalance frequency	weekly
Factors	11 (10 transported, 1 introduced)
Cost model	liquidity-scaled maker fee, 1–5 bp
Funding	subtracted at every rebalance

Second, funding is observable and material. Measured on the deployed book over 282 weeks, the funding term averages -0.173% per week, or -8.99% annualised, and is a net *receipt* in 98% of weeks: because the construction shorts high-funding contracts and holds low-funding ones, the book is a carry collector rather than a payer. A price-only backtest therefore *understates* this strategy by approximately 9% annually. We account for realised funding at every rebalance rather than approximating it.

A resolution constraint discovered by failure. Trade-size distribution factors cannot be computed hourly: a day divides into roughly twelve bars per side, below the twenty-observation minimum the shape estimator requires. Measured computability is 0% at 1-hour and 100% at 5-minute. The remaining intraday factors measured *better* hourly. The pipeline therefore runs at two resolutions simultaneously, a constraint the data imposed, not an optimisation.

4 Methodology

4.1 Factors

Every factor is specified a priori from its source, including its sign. Nothing is estimated. This determines what can be overfit: with no fitted parameters anywhere in the model, the entire overfitting surface reduces to a handful of discrete hyperparameters, and Section 6.3 holds all of them out. Each raw factor is smoothed with a twenty-period rolling mean before ranking.

Table 2 gives definitions. Three have direct academic antecedents: the signed jump measure follows the realised-semivariance decomposition of Barndorff-Nielsen et al. (2010), whose separation of upside from downside variation Patton and Sheppard (2015) show carries predictive content; the order-flow imbalance follows Chordia et al. (2002); and the trade-size distribution factors build on the literature discussed in Section 4.4. The remainder are reproductions of Chinese sell-side research on A-shares. We note plainly that those are practitioner research reports rather than peer-reviewed publications, and are not always formally citable; we reproduce the constructions as described and report their performance without relying on the original claims.

One sign convention is inverted relative to its source. The smart-money factor is documented as a reversal signal in A-shares; in perpetual futures it is momentum-consistent. We report this as an empirical finding rather than a fitted choice, and record it as a limitation: a single inverted sign, chosen with knowledge of the data, is a channel through which information can leak.

Table 2: Factor definitions. All inputs are point-in-time. v_n , n_n and b_n denote volume, trade count and taker-buy volume in intraday bar n ; c_n its close. Weekly quantities are smoothed with a twenty-period rolling mean before ranking.

factor	construction	economic reading	origin
AVOL	$-\log \sum_{12w} \text{volume}$	abnormal turnover predicts negatively	A-share practitioner
Q	VWAP of the top 20% of volume by $ r /\sqrt{v}$, over VWAP of all bars	informed accumulation below the average price	A-share practitioner
RSJ	$-(\sum_{r>0} r^2 - \sum_{r<0} r^2) / \sum r^2$	upside-jump-dominated names underperform	Barndorff-Nielsen et al. (2010)
OFI	$(2 \sum b_n - \sum v_n) / \sum v_n$	net aggressive buying persists	Chordia et al. (2002)
CPVm	$-\text{mean of daily corr}(c_n, v_n)$	price and volume moving together signals crowding	A-share practitioner
CPVv	$-\text{s.d. of daily corr}(c_n, v_n)$	instability in that relation signals regime change	A-share practitioner
WRspread	$-(\text{top-trader L/S} - \text{all-account L/S})$	large accounts crowded relative to the rest	positioning data
TopChg	$-\Delta(\text{top-trader L/S})$	large accounts <i>increasing</i> exposure is a fade	positioning data
Quad	$-\text{sign}(r_w) \cdot \Delta \log(\text{open interest})$	price and open interest rising together is new crowding	CTA practitioner
TKU	kurtosis of $\log(v_n/n_n)$	concentration of the trade-size distribution	A-share practitioner
TSKD	$\Delta[\text{skew}(\log \text{ticket}_B) - \text{skew}(\log \text{ticket}_S)]$	shift in which side places the outsized trades	introduced here

4.2 Directional trade-size asymmetry

Consider one intraday bar containing n_n trades totalling v_n units of volume. Its *ticket size* is v_n/n_n , the typical size of an individual trade within it. Across a day these form a distribution, and the A-share literature establishes that its *shape* carries predictive content that its mean does not.

The limitation of that measure is that a right-skewed distribution is produced equally by a large buyer and a large seller: it detects that unusually sized orders occurred, but not who placed them. With taker-buy volume b_n published per bar, the distribution can be partitioned by initiating side,

$$\mathcal{B} = \{n : b_n > \tfrac{1}{2}v_n\}, \quad \mathcal{S} = \{n : b_n \leq \tfrac{1}{2}v_n\},$$

and its asymmetry measured as

$$\text{TSKD}_t = \Delta \left[\text{skew}(\log \text{ticket}_{n \in \mathcal{B}}) - \text{skew}(\log \text{ticket}_{n \in \mathcal{S}}) \right].$$

A positive value indicates that unusually large trades cluster on the buying side. The *change* carries the signal where the level does not: what predicts is not which side is currently large but which side has *just become* large. The economic motivation is Kyle (1985): an informed trader must transact in size to profit and trades off concealment against execution speed, leaving a detectable footprint rather than none.

The measure requires five-minute bars. At hourly resolution a day divides into roughly twelve bars per side, below the twenty-observation minimum the shape estimator requires; measured computability is 0% at one hour and 100% at five minutes across 80,000 day-records.

4.3 Ranking primitives

Let $f_{i,t}$ denote a raw factor observation for asset i at time t . The cross-sectional score ranks within a date,

$$s_{i,t}^{\text{XS}} = \text{rank}(f_{i,t} \mid \{f_{j,t}\}_{j=1}^N),$$

meaningful only if the $f_{j,t}$ are commensurate. The self-referential score ranks within an asset,

$$s_{i,t}^{\text{TS}} = \text{rank}(f_{i,t} \mid \{f_{i,\tau}\}_{\tau=t-w}^{t-1}),$$

making no cross-asset comparison. Both map to $[-1, +1]$ via $((\text{rank} - 0.5)/n - 0.5) \cdot 2$, which is centred, antisymmetric and NaN-preserving. We use $w = 52$ weeks with a 26-week minimum.

The two degrade differently. As the cross-section becomes more heterogeneous, s^{XS} increasingly ranks *asset identity* rather than *signal*: a contract whose order flow is systematically larger than another's occupies the same tail week after week irrespective of new information. s^{TS} is immune by construction.

4.4 Portfolio formation

Each factor independently forms a dollar-neutral quintile book, long the top 20%, short the bottom 20%, weights $\pm 1/n$. Scores are never blended. Book *returns* are combined by inverse trailing volatility, $w_t^{(k)} \propto 1/\hat{\sigma}_{t-1}^{(k)}$ normalised to sum to one, with the estimate lagged so no contemporaneous

information enters. Where the risk model is not estimable the book is held flat rather than traded equally weighted.

4.5 Carry and turnover

Each score is penalised by the contract’s cross-sectional funding rank, the one place a cross-sectional comparison is appropriate, since funding is denominated identically across contracts. Section 5.3 shows this component is far more consequential than the term “tilt” suggests.

A uniform partial adjustment caps per-rebalance turnover. At the modelled maker fee this *costs* performance monotonically; what it purchases is robustness to cost misestimation. The deployed configuration therefore uses a cap that is not Sharpe-maximising, a deliberate divergence we state rather than leave to be discovered.

4.6 Return construction

$$r_t = \underbrace{w'_t \text{fwd}_t}_{\text{price}} - \underbrace{w'_t \text{funding}_{t+1}}_{\text{carry paid}} - \underbrace{|w_t - w_{t-1}|' c}_{\text{transaction cost}}$$

where c is a liquidity-scaled maker fee from 1 basis point for the most liquid quintile to 5 for the least, interpolated on cross-sectional dollar-volume rank. Forward returns are clipped at +100% so a single listing spike cannot manufacture a Sharpe.

4.7 Point-in-time discipline

Every factor input is leak-tested: a historical score is recomputed with all future observations deleted and required to reproduce exactly. This test exists because it previously *failed*. A ranking helper applied along the wrong array axis ranked each observation against the asset’s entire history, future included, altering historical scores for 99 of 112 contracts (maximum change 1.15). Research and production code agreed perfectly throughout, because both called the same incorrect function. All results here postdate the correction and pass at 0.00e+00.

5 Empirical results

5.1 The comparison frame

Table 3 reports eleven single-factor books under both ranking frames, holding data, universe, costs and construction fixed, both as the strategy trades it and with carry removed entirely.

As traded, mean annualised Sharpe rises from 1.011 to 1.451; an improvement of roughly forty-four per cent obtained without changing a single input, only the set against which each observation is ranked. Nine of eleven factors improve. With carry removed the improvement is larger in

proportional terms and eight of eleven factors improve. **The ranking effect is independent of carry:** the component amplifies the gap but does not create it. No factor changes sign under either specification, so the appropriate claim is degradation of the cross-sectional frame, not its failure. One factor reverses the sign of its ranking preference when carry is removed; at this noise level individual reversals should not be interpreted, and only the aggregate is meaningful.

Table 3: Ranking frame. Eleven single-factor books, annualised Sharpe, net of funding and costs. “Carry removed” sets the funding weight to zero.

specification	cross-sectional	self-referential	gap	self wins
as traded (with carry)	1.011	1.451	+0.440	9 of 11
carry removed	0.383	0.648	+0.265	8 of 11
sign reversals	0 of 11 under either specification			

5.2 Mechanism

Table 4 introduces an intermediate specification: standardise each asset against its own trailing window, then rank cross-sectionally. If scale heterogeneity drives the effect this should recover it. It recovers 41%.

Scale heterogeneity is therefore one channel but not the dominant one. We attribute the residual to distributional robustness: standardisation normalises only the first two moments, whereas a rank against an asset’s own history is invariant to the entire distributional shape. Tests separating the fat-tail component from higher moments are left to future work.

Table 4: Mechanism decomposition.

specification	mean annualised Sharpe	gap recovered
cross-sectional, raw	1.011	,
cross-sectional, per-asset standardised	1.192	41%
self-referential	1.451	100%

5.3 Carry is a factor, not an adjustment

Table 5 reports what removing the funding component costs each book. The effect is large for every factor: the least affected gains roughly forty per cent from its inclusion, the most affected more than triples. Traded on its own the component is a respectable strategy in its own right, though it falls short of the significance threshold we adopt. Carry is therefore a first-order signal in perpetual futures rather than an accounting adjustment. Every single-factor figure in this paper includes it, and we say so rather than presenting those numbers as the ranking effect in isolation.

Table 5: Carry is a factor. Single-factor books with and without the funding component, matched pairwise so each row uses the largest sample valid for that comparison.

factor	with carry	without carry	carry contributes
TSKD	1.442	0.394	+1.048
TKU	1.494	0.674	+0.820
OFI	1.241	0.480	+0.762
AVOL	1.222	0.663	+0.559
Quad	1.896	1.356	+0.540

Notes. Traded on its own over the matched sample, the carry component returns an annualised Sharpe of 1.250 ($t = 2.91$), short of the $t > 3.0$ threshold adopted throughout.

5.4 Construction dominates selection

Table 6 reports the comparison two ways. Panel A is a paired test: every hyperparameter is held fixed and only the construction varies, so the difference is attributable to construction alone. It is large, highly significant ($t = 5.82$), and favours the proposed construction in fifteen of seventeen matched cells. Panel B sweeps all six construction variants over the same grid and ranks them by both their best and their average configuration; the proposed construction places first on both criteria and the conventional pipeline last. The ordering is therefore not an artefact of a single fortunate parameterisation.

Table 6: Construction. Both constructions swept over an identical grid of 1,286 configurations.

Panel A: paired comparison (hyperparameters fixed)		
matched cells	17	
one book per factor, self-referential	1.657	
blended scores, cross-sectional, equal weight	0.992	
mean difference	+ 0.665	
t -statistic of difference	5.82	
cells favouring proposed construction	15 of 17	
Panel B: best-of-grid, all six variants		
construction	best	mean
self / books / risk parity	2.571	1.537
self / blend / equal	2.468	1.288
self / books / equal	2.296	1.510
cross / books / equal	2.200	1.152
cross / books / risk parity	2.113	1.129
cross / blend / equal	1.971	0.995

5.5 Is it alpha?

A Sharpe ratio establishes that a strategy made money; it does not establish that the return is unexplained by factors already known to be priced. We therefore regress the weekly strategy return on three controls constructed from the same panel, an equal-weight market return, a size book formed on trailing dollar volume, and a twelve-week momentum book, each formed point-in-time (Table 7).

Three features stand out. The market loading is -0.001 with $t = -0.12$: the book is market-neutral *empirically*, not merely by construction. Size and momentum load negatively and significantly, but explain little; the regression R^2 is 0.074. And the intercept is large and significant, an annualised alpha of 27.0%, accounting for 97% of the strategy’s mean return.

Because weekly strategy returns are frequently autocorrelated, which biases ordinary t -statistics upward, we report Newey–West standard errors alongside the ordinary ones and quote the former. The correction is modest here, first-order autocorrelation is $+0.063$, moving the alpha t -statistic from 5.03 to 4.42; but it is the conservative figure and it still clears the $t > 3.0$ threshold of Harvey et al. (2016).

Table 7: Factor-adjusted performance. Weekly strategy returns regressed on an equal-weight market factor, a size book and a twelve-week momentum book, all constructed point-in-time from the same panel.

	coefficient	t (OLS)	t (Newey–West)
α	0.00520	5.03	4.42
MKT	-0.00128	-0.13	-0.12
SIZE	-0.09260	-3.83	-2.69
MOM	-0.06819	-3.39	-2.23
annualised α			+27.05%
R^2			0.0739
alpha share of mean return			97%
observations			281 weeks

Notes. Newey–West standard errors use four lags. First-order autocorrelation of the strategy return is $+0.063$. The market coefficient is statistically indistinguishable from zero, confirming empirically the dollar-neutrality imposed by construction.

5.6 Factor contributions

Table 8 reports leave-one-out contributions on a matched 282-week sample, the full book minus each factor, versus the full book. This is the quantity of interest for a portfolio member, as distinct from standalone performance.

The trade-size asymmetry factor ranks **third of eleven on both measures**: third by leave-one-out contribution here, and third by standalone performance on a common sample across all eleven books (1.921 , $t = 3.71$, $n = 194$ weeks, the sample bounded by the three positioning factors). Its

contribution is positive in *both* halves of the sample, indicating structure rather than a single-period artefact. Separately, **four factors carry negative contributions**: the book is marginally better without them. We report the eleven-factor specification because it was fixed a priori; removing the underperformers post hoc would be a selection decision made with knowledge of the outcome, and we decline to make it.

Table 8: Factor contributions. Matched 282-week sample in which every single-factor book is present; full eleven-factor book returns 2.161 ($t = 5.03$).

factor	book without it	contribution
Q	2.033	+0.128
TKU	2.084	+0.078
TSKD	2.115	+0.046
WRspread	2.123	+0.038
AVOL	2.135	+0.026
OFI	2.144	+0.018
TopChg	2.150	+0.011
CPV _v	2.170	−0.008
RSJ	2.179	−0.017
Quad	2.180	−0.019
CPV _m	2.189	−0.028

6 Robustness

6.1 Drawdown, tails, and the 2022 collapse

Statistical significance answers whether a result is real. It does not answer whether the strategy could be held. Table 9 reports the equity curve’s properties and, separately, its behaviour through the cryptocurrency decline of late 2021 and 2022.

Three features are worth drawing out. The maximum peak-to-trough loss is 7.55%, against an annualised return of 27.8%, and it was recovered within sixteen weeks. Skewness is +1.14: the return distribution is right-tailed, which is the opposite of the profile most systematic strategies exhibit and the opposite of the failure mode associated with them. Excess kurtosis of 6.65 confirms that the distribution is far from normal, which is why we report these moments rather than relying on the Sharpe ratio alone.

The second panel tests the neutrality claim where it is most likely to fail. A regression coefficient is an average over a sample; it does not by itself establish behaviour in a crisis. Across the 61 weeks from November 2021 the market fell 82.5% and drew down 84.0%. The strategy returned +9.4% over the same window with a maximum drawdown of 7.6%, and its correlation with the market

across those weeks was +0.045. Neutrality is therefore not only an average property but held through the largest dislocation in the sample.

Table 9: Equity-curve properties and behaviour through the 2021–2022 decline. Panel A covers the full sample; Panel B the 61 weeks from November 2021.

Panel A: full sample (281 weeks)		
cumulative return		+327.6%
annualised return		+27.76%
annualised volatility		12.82%
Sharpe ratio		2.166
maximum drawdown		−7.55%
weeks to recover		16
worst week / worst month	−6.18% / −4.66%	
positive weeks		65.5%
skewness		+1.144
excess kurtosis		+6.651
Panel B: November 2021 to December 2022 (61 weeks)		
	market	strategy
cumulative return	−82.5%	+9.4%
annualised volatility	92.1%	12.6%
Sharpe ratio	−1.11	0.67
maximum drawdown	−84.0%	−7.6%
positive weeks	45.9%	60.7%
correlation with market		+0.045

Notes. The market is the equal-weight return of the tradeable universe. The ratio of annualised return to maximum drawdown is 3.7.

6.2 Walk-forward with rebalance-anchor variation

Configuration is selected on a trailing 104-week window and evaluated on the following 26 weeks, rolling forward. Every hyperparameter is re-selected on each training block, including the quintile fraction: pinning any axis to its full-sample winner would let the test inherit a choice made with knowledge of the evaluation period, which is the precise failure this procedure exists to detect. The candidate set is therefore 36 configurations per fold. Because a weekly strategy also carries an unmeasured exposure to *which weekday* it rebalances (Hoffstein et al. 2019), the panel is rebuilt from hourly bars on each of seven anchors and the procedure repeated (Table 10).

Mean out-of-sample Sharpe is 1.867 and *every one of the seven schedules is profitable*, each on 182 out-of-sample weeks. The dispersion is itself informative: a study reporting a single schedule could have reported anything between 1.074 and 2.575 depending on a choice with no economic content. We report the mean rather than the maximum. Monday, the schedule actually traded,

is second of the seven rather than first, so the traded configuration is not the flattering one. Five of seven anchors clear the Harvey et al. (2016) hurdle of $t > 3.0$; Friday ($t = 2.73$) and Thursday ($t = 2.01$) do not, and we report them as failures rather than excluding them.

The selected quintile fraction varies across folds: 0.10, 0.20 and 0.30 are each chosen in at least one block, and the single best-performing fold selects 0.10. No one value drives the result, which is the substantive reason the parameter can be treated as incidental rather than load-bearing.

Table 10: Walk-forward out-of-sample Sharpe by rebalance anchor. Every hyperparameter, including the quintile fraction, is selected within each training block; 36 candidates per fold, 182 out-of-sample weeks per anchor. Harvey et al. (2016) hurdle is $t > 3.0$.

anchor	Sun	Mon	Wed	Sat	Tue	Fri	Thu	mean
OOS Sharpe	2.575	2.541	1.885	1.840	1.696	1.460	1.074	1.867
t	4.82	4.75	3.53	3.44	3.17	2.73	2.01	
clears $t > 3$	yes	yes	yes	yes	yes	no	no	5 of 7

Notes. Walk-forward efficiency, out-of-sample Sharpe as a percentage of the in-sample best, ranges from 61% (Thursday) to 113% (Sunday), above the 50% threshold on every anchor. An earlier version of this test held the quintile fraction fixed at its full-sample value and reported a mean of 2.068; releasing that constraint lowers the figure by 0.20 and costs two anchors their $t > 3$ status. The lower number is the correct one.

6.3 Fold geometry

Table 11 varies the train and test window lengths across seven configurations. Every one produces a profitable out-of-sample series and every one clears the Harvey threshold, on out-of-sample Sharpe ratios spanning 1.998 to 2.731. There is no monotone relationship between training-window length and performance: the longest window tested is mid-range and the second-shortest is near the top. We read this as the result being insensitive to fold geometry rather than as evidence about the optimal window, and the geometry used for the headline result is mid-range rather than maximal.

Table 11: Fold geometry. Monday anchor; only the train/test split varies. All hyperparameters, including the quintile fraction, are re-selected within each training block.

train	test	folds	n_{oos}	OOS Sharpe	t
78	13	16	208	2.609	5.22
78	26	8	208	2.525	5.05
104	13	14	182	2.578	4.82
104	26	7	182	2.541	4.75
104	52	3	156	2.731	4.73
130	26	6	156	1.998	3.46
156	26	5	130	2.690	4.25

6.4 Held-out universe

Time-series testing does not establish that hyperparameters generalise across *assets*. We hold out symbols: configuration is selected on 78 contracts and evaluated on 34 that had no influence on the choice, across eight random splits, with the cross-section rebuilt within each side so ranks, quintile boundaries and the cost model are computed only among symbols present.

A naive comparison is misleading here, and we flag it because we initially made it. A 34-contract book is mechanically inferior to a 78-contract one, since information ratio scales with the square root of breadth (Grinold 1989). Table 12 therefore adds a matched-breadth control: the same selected configuration evaluated on a random 34 of the *training* contracts.

All eight splits are profitable out of universe. More informative than the ratio is the stability of the selection: all eight chose identical funding weight, turnover treatment and quintile width, and five of eight the same ranking window; fitted noise would not converge this way. We note that only one of eight held-out splits reaches $t > 3.0$, a consequence of reduced breadth, and that the held-out and control distributions overlap substantially. The defensible claim is that retention is materially above 50%, not precisely 83%.

Table 12: Held-out universe, eight random splits. The matched-breadth control evaluates the same selected configuration on a random 34 of the *training* contracts, isolating overfitting from the mechanical breadth effect.

	train (78)	held-out (34)	control: seen (34)
mean annualised Sharpe	2.204	1.127	1.366
median	2.169	1.194	1.407
standard deviation	0.212	0.286	0.281
splits profitable	,	8 of 8	,
held-out / train	51%	<i>conflates two effects; not interpretable</i>	
control / train	62%	<i>breadth, mechanical</i>	
held-out / control	83%	overfitting, at matched breadth	

6.5 Multiple testing with a measured trial count

The deflated Sharpe ratio (Bailey and López de Prado 2014) requires the number of trials and their dispersion. Both describe the *search* rather than the strategy and cannot be recovered from the winning backtest, so in practice they are supplied from recollection, precisely the quantity the correction exists to discipline. We instead log every evaluated configuration to an append-only registry, written by the code performing the evaluation, and read both inputs from it (Table 13).

We report both readings. The defensible choice is $N = 211$, since roughly 1,075 of the logged cells constitute the construction ablation executed *after* the strategy was specified, to measure the baseline fairly rather than to select among candidates. The conservative reading falls below 0.95

and we state it. Its instability is itself the argument against it: the value fell from 0.9507 to 0.9475 purely because an ablation continued running. A statistic that penalises conducting additional controlled experiments is not measuring selection bias.

Table 13: Deflated Sharpe ratio. Trial count and dispersion read from an executed experiment registry rather than supplied by the researcher.

trial set	N	sd(SR)	SR^*	DSR
strategy-search configurations	211	0.0768	0.2139	0.9885
all logged configurations	1,286	0.0766	0.2547	0.9475

Notes. The selected strategy has $T = 269$ weekly observations, a weekly Sharpe of 0.3565, skewness +0.145 and kurtosis 4.391. Against a zero benchmark the probabilistic Sharpe ratio is 1.0000. SR^* is the expected maximum Sharpe across N zero-skill trials.

6.6 Transaction costs

Table 14 scales the entire liquidity-adjusted cost curve by successive factors of two. The uncapped book is superior only while realised costs remain close to the modelled level; beyond roughly four times it the ordering reverses, and by the extreme of the range the uncapped book loses money while the deployed configuration remains profitable. This is the sense in which the turnover cap is insurance rather than alpha.

Table 14: Transaction-cost robustness. The entire liquidity-scaled cost curve is multiplied by the stated factor.

cost multiple	1×	2×	4×	8×	16×	32×
uncapped	2.364	2.247	2.014	1.547	0.618	−1.222
deployed (capped)	2.161	2.117	2.028	1.851	1.496	0.789

6.7 Research–production equivalence

The relationship between a published backtest and the system trading it is normally asserted. We test it. Research and live paths call a single shared implementation, and an automated gate asserts point-in-time integrity, determinism, dollar-neutrality, non-emptiness and identical position directions, exiting non-zero so it can block deployment (Table 15).

7 Economic significance and capacity

A Sharpe ratio is not an investment. Table 16 estimates deployable capital from realised liquidity, the book’s measured breadth (76 concurrent positions) and measured weekly turnover (0.369 of gross), under a participation constraint on the 25th-percentile contract by average daily dollar

Table 15: Research–production equivalence gate. Exits non-zero on any failure.

assertion	measured	status
point-in-time leak, all factors	0.00e+00	pass
determinism	exact	pass
dollar-neutrality (max net)	9.02e−17	pass
gross exposure bound	$1.462 \leq 2.0$	pass
non-emptiness	308 built, 282 active	pass
research = live directions	0 / 4,480	pass

volume, the binding constraint, since the book must trade its illiquid members as well as its liquid ones.

The implied figures run from tens to hundreds of millions of dollars depending on the participation rate assumed. These are order-of-magnitude bounds rather than precise limits: they assume participation scales linearly with capital and ignore that market impact is convex in size, so the true ceiling is lower than the linear extrapolation suggests. They also inherit the maker-fee assumption, which Section 6.5 shows the deployed configuration survives at eight times the modelled level, bounding, rather than resolving, the concern.

Table 16: Capacity. Binding constraint is the 25th-percentile contract by average daily dollar volume (\$3.26m); breadth 76 positions, weekly turnover 0.369.

participation rate	implied capacity
1%	\$33.6m
2%	\$67.2m
5%	\$167.9m
10%	\$335.8m

Notes. Median contract average daily dollar volume is \$6.37m; the tenth percentile is \$1.93m. The binding constraint is the 25th percentile at \$3.26m.

8 What did not survive

8.1 An audit of our own claims

Before submission we searched the research codebase for the executable artifact behind every headline figure. Four had none (Table 17). A separate claim, that one factor reverses sign between ranking frames, was tested directly and failed: zero of eleven reverse.

We report this because the corrected figures appear in this paper and the discarded ones do not, and because the episode motivates the infrastructure in Sections 6.4 and 6.6.

Table 17: Claims that did not survive audit. All are excluded from this paper.

claim as previously recorded	what the audit found
walk-forward Sharpe “2.68–2.95”	tail-index statistics from a document concerning an unrelated strategy; not Sharpe ratios
timing-luck dispersion “ ± 0.24 ”	no script, output or artifact
deflated Sharpe “0.9966 / 0.9816”	no artifact
conventional baseline “1.03”	no artifact; approximately the conventional construction’s <i>mean</i> (0.992) compared against the proposed construction’s <i>maximum</i>
one factor reverses sign between frames	tested directly; zero of eleven reverse

8.2 Negative results

We conjectured that crypto factors are structurally more orthogonal than equity factors, which would make combining them mechanically more valuable here than in equities. Measured mean pairwise correlation among Alpha101-style books in this market is +0.2173 across 4,560 factor pairs. Kakushadze (2016), who introduced these formulaic alphas, reports an average pairwise correlation of 15.9% for the same construction in equities. The conjecture is therefore rejected: factors here are *more* correlated, not less, by roughly a third. The orthogonality that makes per-factor construction valuable derives from data-source diversity, price, flow, positioning, trade-size distribution, not from the asset class.

We report Pearson information coefficients alongside Spearman ones throughout. Returns in this market are severely fat-tailed, and the two can diverge: a factor may order the typical asset correctly while being wrong about the small number of observations that dominate realised profit. Rank correlation measures the former, which is not what a portfolio is paid on. We also report decile monotonicity, since a genuine cross-sectional sort has a gradient and a positive average information coefficient alone does not establish one. We report Pearson alongside Spearman throughout.

9 Conclusion

Factor investing imported into cryptocurrency perpetual futures inherits an assumption this market does not support: that its constituents are comparable. We find the assumption costly but not fatal. Cross-sectional ranking remains profitable, while ranking each contract against its own history performs materially better in nine of eleven cases, eight of eleven with carry removed: for reasons we partly decompose. We further find that how signals are combined after ranking matters more than any individual signal, and we exhibit a factor whose construction depends on information this market publishes exactly and equity research must infer.

We emphasise the structure of the evidence as much as any single result: out-of-sample in time and across assets, replicated over seven rebalance schedules and seven fold geometries, corrected for multiple testing using a trial count read from an executed log, stress-tested to eight times modelled costs, and verified to be computed identically by the deployed system. Each test was capable of overturning the result.

Limitations. Our evidence covers one venue over roughly seven years. Out-of-sample testing is conducted in time and across the cross-section, but not across venues. Capacity is estimated under a linear participation assumption that understates convex impact. One factor’s sign convention was inverted with knowledge of the data. The held-out universe result rests on eight splits. Four of the eleven factors in our sample contribute negatively on leave-one-out; we retain them because the specification was fixed before measurement, but a practitioner building a book rather than testing a method would drop them.

Data and code availability

All results in this paper are reproducible from publicly available data. Price, volume and funding series are drawn from a public exchange bulk archive; positioning series from a keyless public metrics endpoint. No proprietary or paid data is used. The analysis code, including the backtest, the walk-forward and held-out-universe procedures, the experiment registry from which the multiple-testing correction reads its inputs, and the research–production equivalence gate, is available at https://github.com/dhanyamd/mlops/tree/main/quant_signal. The replication set is collected under [research/code/](#), whose README maps each script to the table in this paper that it produces.

Appendix A: Summary of statistical evidence

Table 18: Summary of statistical evidence. Every test reported in the paper, the threshold it is judged against, and the outcome. Three tests do not clear their threshold and are marked accordingly.

test	statistic	threshold	outcome
In-sample significance			
t -statistic, full book	$t = 5.03, n = 282$	$t > 3.0$ (Harvey)	pass
Probabilistic Sharpe vs zero	PSR = 1.0000	> 0.95	pass
Multiple-testing correction			
Deflated Sharpe, strategy search	0.9885, $N = 211$	> 0.95 (Bailey)	pass
Deflated Sharpe, all logged cells	0.9475, $N = 1,286$	> 0.95	fail
Out-of-sample in time			
Walk-forward, anchor-averaged	$SR = 1.867$	> 0	pass
Walk-forward, per anchor	$t = 2.01\text{--}4.82$	$t > 3.0$	5 of 7
Walk-Forward Efficiency	103%	$> 50\text{--}60\%$	pass
Fold-geometry robustness	$t = 3.46\text{--}5.22$	$t > 3.0$	7 of 7
Out-of-sample across assets			
Held-out universe, 8 splits	8 of 8 profitable	> 0	pass
Retention at matched breadth	83%	$> 50\%$	pass
Held-out per-split t	1 of 8 reach $t > 3.0$	$t > 3.0$	1 of 8
Rebalance-schedule luck (Hoffstein)			
Timing luck, 7 anchors in-sample	$t = 3.67\text{--}5.03$	$t > 3.0$	7 of 7
Anchors profitable	7 of 7	,	pass
Construction effect			
Paired difference, 17 cells	$+0.665, t = 5.82$	$t > 3.0$	pass
Cells favouring construction	15 of 17	$> 50\%$	pass
Individual components			
Trade-size asymmetry, standalone	$t = 3.35$	$t > 3.0$	pass
Trade-size asymmetry, subperiods	$+0.030 / +0.079$	both > 0	pass
Carry, standalone	$t = 2.91$	$t > 3.0$	fail
Cost and capacity			
Cost robustness, deployed book	viable to $8\times$	$\geq 1\times$	pass
Capacity at 2% participation	\$67m	,	,
Implementation integrity			
Point-in-time leak, all factors	0.00e+00	$= 0$	pass
Dollar-neutrality	9.02e−17	≈ 0	pass
Research = live directions	0 of 4,480	$= 0$	pass
Determinism	exact	exact	pass

Appendix B: Rejected constructions

The multiple-testing correction in Section 6.4 reads its trial count from an executed log. Consistency requires that the constructions abandoned during development be disclosed rather than omitted; a search branch that failed is still a search branch. Nineteen are enumerated below. All were built, measured and rejected before the specification reported in this paper was fixed.

Panel A: signal constructions (12)

construction	what it was	why rejected
FAS	funding accrual residualised on the price path	measured IC -0.001 (Pearson), $+0.002$ (Spearman); $ t < 0.4$, indistinguishable from zero
RCGO	capital-gains overhang adapted to crypto	degraded every configuration tested
Salience / ST	salience-theory weighting of return distributions	Spearman IC significant, Pearson not; no decile gradient
SAT	salience-adjusted turnover	no incremental contribution
IFD	smart-money bar selection made sign-aware	subsumed by simpler order-flow imbalance
Kyle asymmetry	price impact per unit buy vs sell volume	unstable; requires per-side impact estimates the sample cannot support
session-RPV	price-volume correlation split by global session	sessions do not partition crypto flow as trading hours do equities
LOP	limit-order pressure proxy	not constructible from public bar data
WCC	weighted crowding composite	collinear with existing positioning factors
FSD	flow-shape divergence	superseded by the trade-size asymmetry factor
TCN-ACN	top-count minus all-count normalisation	equivalent to an existing positioning factor after rescaling
forward funding cost	expected rather than realised funding	forecast error exceeded the signal

Panel B: portfolio constructions (7)

construction	what it was	why rejected
LSN	latent-sector neutralisation via principal components, with the component count set by the Marchenko–Pastur noise edge	removed signal along with the shared mode; no net gain
priority turnover	spend the turnover budget on the largest signal changes first	destroyed the uniform-adjustment property that makes the cap work
cost-aware turnover	vary adjustment speed by estimated cost	same failure mode as above
clienteles conditioning	condition weights on positioning composition	no incremental contribution
leg specialisation	different construction for the long and short legs	no incremental contribution
MOD-correction transfer	transfer a published mispricing correction	does not survive the change of market
orthogonalisation transfer	symmetric orthogonalisation of the factor matrix before ranking	no gain once returns rather than scores are combined

Pre-submission checklist

- Cite and differentiate the Management Science crypto factor-pricing literature; Section 5.1 is a direct empirical response.
- Insert the AI-use disclosure with the abstract (SSRN requirement).
- Data and code availability statement (Management Science requirement).
- Anonymise for Management Science: remove name, affiliation, acknowledgements; strip PDF metadata.
- Appendix: nineteen rejected constructions, enumerated.
- Appendix: 100-word plain-language abstract for the practitioner version.

References

- Asness CS, Moskowitz TJ, Pedersen LH (2013) Value and momentum everywhere. *Journal of Finance* 68(3):929–985.
- Bailey DH, López de Prado M (2014) The deflated Sharpe ratio. *Journal of Portfolio Management* 40(5):94–107.
- Barndorff-Nielsen OE, Kinnebrock S, Shephard N (2010) Measuring downside risk: Realised semi-variance. In *Volatility and Time Series Econometrics* (Oxford University Press), 117–136.
- Chordia T, Roll R, Subrahmanyam A (2002) Order imbalance, liquidity, and market returns. *Journal of Financial Economics* 65(1):111–130.
- Cong LW, Karolyi GA, Tang K, Zhao W (2026) Crypto value, factor pricing, and market segmentation. *Management Science*, Articles in Advance. <https://doi.org/10.1287/mnsc.2024.05875>.
- Daniel K, Titman S (1997) Evidence on the characteristics of cross sectional variation in stock returns. *Journal of Finance* 52(1):1–33.
- Fama EF, French KR (1993) Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33(1):3–56.
- Grinold RC (1989) The fundamental law of active management. *Journal of Portfolio Management* 15(3):30–37.
- Gu S, Kelly B, Xiu D (2020) Empirical asset pricing via machine learning. *Review of Financial Studies* 33(5):2223–2273.
- Harvey CR, Liu Y, Zhu H (2016) ... and the cross-section of expected returns. *Review of Financial Studies* 29(1):5–68.
- Hoffstein C, Sibears J, Faber N (2019) Rebalance timing luck: The difference between hired and fired. *Journal of Index Investing* 10(1):27–36.
- Kakushadze Z (2016) 101 formulaic alphas. *Wilmott* 2016(84):72–81. <https://doi.org/10.1002/wilm.10525>.
- Kyle AS (1985) Continuous auctions and insider trading. *Econometrica* 53(6):1315–1335.
- Lee CMC, Ready MJ (1991) Inferring trade direction from intraday data. *Journal of Finance* 46(2):733–746.
- Moskowitz TJ, Ooi YH, Pedersen LH (2012) Time series momentum. *Journal of Financial Economics* 104(2):228–250.

Patton AJ, Sheppard K (2015) Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Review of Economics and Statistics* 97(3):683–697.

Zhou WX (2008) Universal price impact functions of individual trades in an order-driven market. *Quantitative Finance*. <https://arxiv.org/abs/0708.3198>.